

1 "

$$2. \varphi_1 = x(1-x)$$

$$\varphi_2 = x(1-x)(2x-1)$$

$$u_a = c_1 x(1-x) + c_2 x(1-x)(2x-1)$$

$$= c_1 (-x^2 + x) + c_2 x(2x-1-2x^2+x)$$

$$= \quad \quad \quad + c_2 (-2x^3 + 3x^2 - x)$$

$$u_a' = c_1 (-2x+1) + c_2 (-6x^2+6x-1)$$

$$A \cdot c = b \quad \leftarrow L(\varphi_i)$$

$$\begin{pmatrix} a(\varphi_1, \varphi_1) & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} L(\varphi_1) \\ L(\varphi_2) \end{pmatrix}$$

3.

$$a(\varphi_1, \varphi_1) = 11/30$$

$$a(\varphi_1, \varphi_2) = \int_0^1 (1-2x)(-6x^2+6x-1)dx + \int_0^1 x(1-x) \overset{(x^4-2x^3+x^2)(2x-1)}{x(1-x)(2x-1)} dx$$

$$= \int_0^1 12x^3 - 18x^2 + 8x - 1 dx + \int_0^1 2x^5 - 5x^4 - x^2 dx$$

$$= \left[3x^4 - 6x^3 + 4x^2 - x \right]_0^1 + \left[\frac{1}{3}x^6 - x^5 - \frac{1}{3}x^3 \right]_0^1$$

$$= 0 + \frac{1}{3} - 1 - \frac{1}{3} = -1$$

$$a(\varphi_2, \varphi_2) = \int_0^1 (-6x^2+6x-1)(-6x^2+6x-1)dx + \int_0^1 x(1-x)(2x-1)x(1-x)(2x-1)dx$$

$$= \int_0^1 36x^4 - 72x^3 + 48x^2 - 12x + 1 dx + \int_0^1 4x^6 - 12x^5 + 13x^4 - 6x^3 + x^2 dx$$

$$= \left[\frac{36}{5}x^5 - 18x^4 + \frac{48}{3}x^3 - 6x^2 + x \right]_0^1 + \left[\frac{4}{7}x^7 - 2x^6 + \frac{13}{5}x^5 - \frac{6}{4}x^4 + \frac{1}{3}x^3 \right]_0^1$$

$$= 1/210$$

4.

$$L(\varphi_1) = 1/12$$

$$L(\varphi_2) = \int_0^1 x x(1-x)(2x-1) dx = \int_0^1 -2x^4 + 3x^3 - x^2 dx = \left[-\frac{2}{5}x^5 + \frac{3}{4}x^4 - \frac{1}{3}x^3 \right]_0^1 = \frac{1}{60}$$

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$$\begin{pmatrix} 11/30 & -1 \\ -1 & 1/210 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 1/12 \\ 1/60 \end{pmatrix} \quad \begin{array}{l} \text{I} \\ \text{II} \end{array} \quad \begin{array}{l} 11/30 c_1 - c_2 = 1/12 \\ -c_1 + 1/210 c_2 = 1/60 \end{array} \Rightarrow \begin{array}{l} c_1 = -0,0171 \\ c_2 = -0,0896 \end{array}$$

$$\tilde{u}(x) = -0,0171x(1-x) - 0,0896x(1-x)(2x-1)$$